

Manhattan Vacation Rental Market Analysis

Project Overview

In this project, I analyzed Airbnb rental data in Manhattan to identify which neighborhoods and property sizes are most attractive for vacation rentals. The goal was to help a client understand where potential investment opportunities may be strongest based on rental demand, customer activity, and estimated revenue.

Using Google Sheets, pivot tables, and spreadsheet functions, I cleaned and analyzed the data to uncover trends in neighborhood popularity, bedroom preferences, and earning potential.

Business Questions

This analysis focused on answering the following questions:

1. Which Manhattan neighborhoods are most attractive for vacation rentals?
2. What property sizes are most popular among renters?
3. Do bedroom preferences vary across neighborhoods?
4. How much revenue could top-performing listings potentially generate?

Tools Used

1. Google Sheets
2. Pivot Tables
3. Spreadsheet Functions
4. Data Cleaning
5. Charts & Visualizations

Data Cleaning

Before beginning the analysis, I cleaned several columns to improve consistency and accuracy across the dataset.

Cleaning Steps

1. Removed inconsistent capitalization and extra spaces from neighborhood names
2. Created a cleaned column called `neighborhood\_clean`
3. Replaced empty bedroom values with `0` to represent studio apartments
4. Created a new column called `bedrooms\_clean`
5. Added calculated revenue columns using spreadsheet formulas

These steps helped ensure the pivot tables and revenue calculations were accurate and reliable.

Analysis & Findings

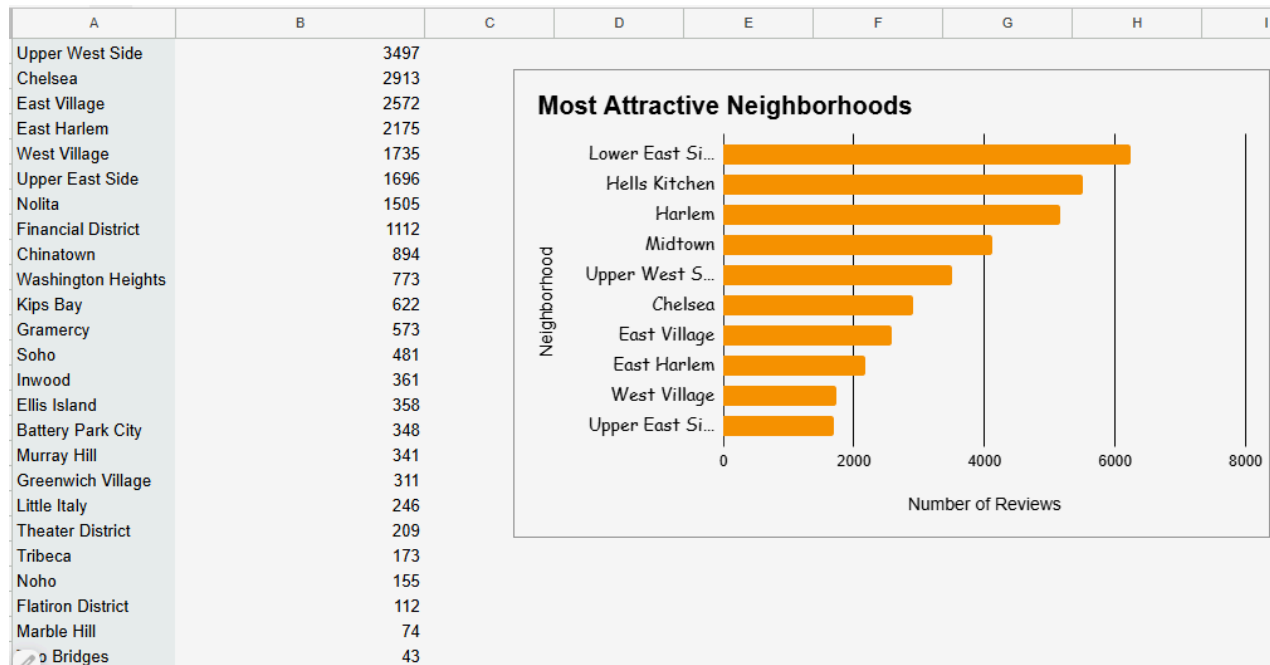
Neighborhood Analysis

I used the `number\_of\_reviews\_ltm` column as an indicator of rental activity and demand. Based on the analysis, some of the most attractive neighborhoods included:

1. Harlem
2. East Village
3. Upper West Side

4. Midtown
5. Lower East Side

A bar chart was created to compare review activity across the top-performing neighborhoods.



Property Size Analysis

I analyzed bedroom counts to determine which property sizes were most popular among renters.

Key Findings

1. 1-bedroom apartments were the most common and most popular property type
2. Studio apartments performed especially well in Midtown
3. Different neighborhoods showed different preferences for apartment size

A	B	C	D	E	F	G	H	I	J	K
COUNT of ic	bedrooms									
neighborhood_c	0	1	2	3	4	5	6	7	8	Grand Total
Upper East Side	33	75	18	5	2	1			1	135
West Village	10	53	16	10	2					91
Washington Heig	3	29	11	4	2					49
Upper West Side	22	93	55	22	2	2				196
Two Bridges	1	3		1						5
Tribeca		8	4	1	0					13
Theater District	2	5	5							12
Stuyvesant Town		2		1						3
Soho	6	17	12	3						38
Roosevelt Island		1								1
Nolita	5	43	7	1						56
Noho		8	1	1						10
Murray Hill	3	9	2	0						14
Morningside Heights		1	1							2
Midtown	84	66	31	4	2					187
Marble Hill			2	1						3
Lower East Side	11	202	23	1	0		1			238
Little Italy		6	5	1						12
Kips Bay	4	17	7	0	1					29
Inwood	2	15	5	1						23
Hells Kitchen	23	105	70	16	3			1		218
Harlem	48	132	65	25	11	5				286
Greenwich Villag	5	13	4	0	0					22

Revenue Analysis

To estimate earnings, I used the calendar dataset to calculate how much revenue listings generated over a 30-day period and projected annual revenue.

Revenue Process

1. Created a `revenue\_earned` column in the calendar sheet
2. Counted revenue only for dates when listings were rented
3. Used `SUMIF()` to calculate total earnings for each listing
4. Estimated annual revenue by multiplying 30-day earnings by 12

Example Formula

excel

=SUMIF(calendar!\$A\$2:\$A\$72554,A2,calendar!\$H\$2:\$H\$72554)

listing_id	date	available	price	adjusted_price	minimum_nights	maximum_nights	revenue_earned
9357	2022-09-07	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-08	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-09	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-10	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-11	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-12	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-13	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-14	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-15	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-16	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-17	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-18	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-19	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-20	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-21	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-22	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-23	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-24	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-25	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-26	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-27	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-28	f	\$175.00	\$175.00	30	1125	\$175.00
9357	2022-09-29	f	\$175.00	\$175.00	30	1125	\$175.00

Key Insights

1. Harlem and East Village showed strong rental demand based on review activity
1-bedroom apartments appeared to be the strongest overall investment option
2. Midtown showed higher demand for studio apartments
3. Some listings demonstrated strong annual revenue potential based on occupancy and pricing

Recommendations

Based on the analysis:

1. Investors may benefit from focusing on 1-bedroom apartments in high-demand neighborhoods
2. Midtown may be better suited for studio apartment investments
3. Neighborhoods with higher review activity may indicate stronger rental demand and occupancy rates

Skills Demonstrated:

1. Spreadsheet Analysis
2. Data Cleaning
3. Pivot Tables
4. Revenue Analysis
5. Business Reporting
6. Data Visualization
7. Analytical Thinking